

Практика 3.2

Остановка контейнеров Docker Compose командой `docker compose down`.

```
Microsoft Windows [Version 10.0.19045.6456]
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C:\Users\Malika\Desktop\microservices-demo>dir
Том в устройстве C не имеет метки.
Серийный номер тома: 7A77-58A6

Содержимое папки C:\Users\Malika\Desktop\microservices-demo

09.04.2026 17:39 <DIR>      .
09.04.2026 17:39 <DIR>      ..
09.04.2026 16:57          390 docker-compose.yml
09.04.2026 17:43 <DIR>      order-service
09.04.2026 17:44 <DIR>      user-service
                1 файлов      390 байт
                4 папок 184 703 582 208 байт свободно

C:\Users\Malika\Desktop\microservices-demo>docker compose down
[+] down 3/3
0.0s Container order-service Removed
0.0s Container user-service Removed
0.0s Network microservices-demo_micro-net Removed
```

Инициализация Docker Swarm командой `docker swarm init`.

Проверка узлов кластера Docker Swarm командой `docker node ls`.

```
C:\Users\Malika\Desktop\microservices-demo>docker swarm init
Swarm initialized: current node (bmx9l75j6sgrmqz9nryyq4c4s) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-000h4y7shbljeau0rk9b8ktj26qxjqlnp7k0nulk4c1mhybhly-cl1c8lmhbtb850vb7zyjxtxvg 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.

C:\Users\Malika\Desktop\microservices-demo>docker node ls
ID                HOSTNAME        STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
bmx9l75j6sgrmqz9nryyq4c4s * docker-desktop Ready        Active           Leader             29.3.1
```

Создание overlay-сети Docker Swarm командой `docker network create -d overlay swarm-net`.

Сборка Docker-образа микросервиса `user-service`.

```
C:\Users\Malika\Desktop\microservices-demo>docker network create -d overlay swarm-net
93le3lq83g7fczubvkhnt124

C:\Users\Malika\Desktop\microservices-demo>docker build -t microservices-demo-user-service ./user-service
[+] Building 1.9s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                docker:desktop-linux 0.0s
=> => transferring dockerfile: 214B                                              0.0s
=> [internal] load metadata for docker.io/library/python:3.11-alpine             1.6s
=> [internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                    0.0s
=> [1/5] FROM docker.io/library/python:3.11-alpine@sha256:f07e2ace46f560f09a6eeec7b4913b80ee99546e749ef82342a419 0.0s
=> => resolve docker.io/library/python:3.11-alpine@sha256:f07e2ace46f560f09a6eeec7b4913b80ee99546e749ef82342a419 0.0s
=> [internal] load build context                                                  0.0s
=> => transferring context: 123B                                                  0.0s
=> CACHED [2/5] WORKDIR /app                                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                                           0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                0.0s
=> CACHED [5/5] COPY . .                                                         0.0s
=> exporting to image                                                             0.2s
=> => exporting layers                                                            0.0s
=> => exporting manifest sha256:819b777c04d16ab58cbb9cc217baf531a2f8d3162842816b3c74460633344397 0.0s
=> => exporting config sha256:63f39324aca748ac9da455908873cadeb81c976b079d4c97bf6516f508500aca 0.0s
=> => exporting attestation manifest sha256:776b800c8c558c9011d78dff3606b5f7755ea165109e46bfe5e89025db3311e4 0.0s
=> => exporting manifest list sha256:0bc9d8144daf2a1cf0920c5c09d07f1da6123b9f0aac787ffbe55d10aa96cd02 0.0s
=> => naming to docker.io/library/microservices-demo-user-service:latest         0.0s
=> => unpacking to docker.io/library/microservices-demo-user-service:latest      0.0s
```

Сборка Docker-образа микросервиса order-service.

```
C:\Users\Malika\Desktop\microservices-demo>docker build -t microservices-demo-order-service ./order-service
[+] Building 0.6s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 214B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-alpine 0.3s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                      0.0s
=> [1/5] FROM docker.io/library/python:3.11-alpine@sha256:f07e2ace46f560f09a6eeec7b4913b80ee99546e749ef82342a419 0.0s
=> => resolve docker.io/library/python:3.11-alpine@sha256:f07e2ace46f560f09a6eeec7b4913b80ee99546e749ef82342a419 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 124B                                    0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/5] COPY . .                                           0.0s
=> exporting to image                                              0.1s
=> => exporting layers                                              0.0s
=> => exporting manifest sha256:c265d6f771db157d149b7b312f561537023b506450a93d86ce8d97e0afc6d0df 0.0s
=> => exporting config sha256:34298af05c8612cb97e6ad6153e9bb7cddac8a1945c03375e1f9efe0a40fc9a1 0.0s
=> => exporting attestation manifest sha256:8779c8fe398c44166a5c782c84ad619b60256ea0da29a5f57b490520e745776e 0.0s
=> => exporting manifest list sha256:e48d7aa45f1688275d4ce82313473c7e36396413d1479ca69cc60b4c72863b38 0.0s
=> => naming to docker.io/library/microservices-demo-order-service:latest 0.0s
=> => unpacking to docker.io/library/microservices-demo-order-service:latest 0.0s
```

Развёртывание сервиса user-service в Docker Swarm.

```
C:\Users\Malika\Desktop\microservices-demo>docker service create --name user-service --network swarm-net -p 3001:3001 microservices-demo-user-service
image microservices-demo-user-service:latest could not be accessed on a registry to record
its digest. Each node will access microservices-demo-user-service:latest independently,
possibly leading to different nodes running different
versions of the image.

3wd5uzl19jhj0e2ri274mf7hf
overall progress: 1 out of 1 tasks
1/1: running [=====>]
verify: Service 3wd5uzl19jhj0e2ri274mf7hf converged
```

Развёртывание сервиса order-service в Docker Swarm.

```
C:\Users\Malika\Desktop\microservices-demo>docker service create --name order-service --network swarm-net -p 3002:3002 microservices-demo-order-service
image microservices-demo-order-service:latest could not be accessed on a registry to record
its digest. Each node will access microservices-demo-order-service:latest independently,
possibly leading to different nodes running different
versions of the image.

uk3tled9grr57fr2bwnbo58i0
overall progress: 1 out of 1 tasks
1/1: running [=====>]
verify: Service uk3tled9grr57fr2bwnbo58i0 converged
```

Проверка списка запущенных сервисов Docker Swarm.

```
C:\Users\Malika\Desktop\microservices-demo>docker service ls
ID            NAME          MODE         REPLICAS  IMAGE                                  PORTS
uk3tled9grr5  order-service replicated   1/1        microservices-demo-order-service:latest *:3002->3002/tcp
3wd5uzl19jhj  user-service  replicated   1/1        microservices-demo-user-service:latest *:3001->3001/tcp
```

Проверка состояния задач сервиса user-service и order

```
C:\Users\Malika\Desktop\microservices-demo>docker service ps user-service
ID            NAME          IMAGE                                NODE           DESIRED STATE  CURRENT STATE          ERROR          PORTS
ykg22g0mzxf6  user-service.1 microservices-demo-user-service:latest docker-desktop Running         Running about a minute ago
C:\Users\Malika\Desktop\microservices-demo>docker service ps order-service
ID            NAME          IMAGE                                NODE           DESIRED STATE  CURRENT STATE          ERROR          PORTS
x9tkn8xvf2tw  order-service.1 microservices-demo-order-service:latest docker-desktop Running         Running about a minute ago
```

Проверка работы сервисов после развёртывания в Docker Swarm.

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3001/users
[{"email":"alice@example.com","id":1,"name":"Alice"}, {"email":"bob@example.com","id":2,"name":"Bob"}]

C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3002/orders
[{"amount":1,"id":1,"item":"Laptop","user_id":1}, {"amount":2,"id":2,"item":"Mouse","user_id":2}]
```

Создание конфигурационного файла NGINX для проксирования запросов к микросервисам.

nginx.conf – Блокнот

Файл Правка Формат Вид Справка

```
events {
    worker_connections 1024;
}

http {
    upstream user_service {
        server user-service:3001;
    }

    upstream order_service {
        server order-service:3002;
    }

    server {
        listen 80;

        location /users {
            proxy_pass http://user_service;
            proxy_set_header Host $host;
        }

        location /orders {
            proxy_pass http://order_service;
            proxy_set_header Host $host;
        }
    }
}
```

проверка

```
C:\Users\Malika\Desktop\microservices-demo>dir
Том в устройстве C не имеет метки.
Серийный номер тома: 7A77-58A6

Содержимое папки C:\Users\Malika\Desktop\microservices-demo

10.04.2026  15:54    <DIR>          .
10.04.2026  15:54    <DIR>          ..
09.04.2026  16:57             390 docker-compose.yml
10.04.2026  15:55             499 nginx.conf
09.04.2026  17:43    <DIR>          order-service
09.04.2026  17:44    <DIR>          user-service
                2 файлов                889 байт
                4 папок   184 701 370 368 байт свободно
```

Развёртывание NGINX в Docker Swarm командой

```
C:\Users\Malika\Desktop\microservices-demo>docker service create --name nginx --network swarm-net -p 80:80 --mount type=bind,source=%cd%\nginx.conf,target=/etc/nginx/nginx.conf nginx:alpine
nginx:alpine: service is: 1 out of 1 tasks
/1: running [=====]
Verify: Service nujpcc93ug8ujoe6megFbfyr converged
```

проверка через nginx

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost/users
[{"email":"alice@example.com","id":1,"name":"Alice"}, {"email":"bob@example.com","id":2,"name":"Bob"}]

C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost/orders
[{"amount":1,"id":1,"item":"Laptop","user_id":1}, {"amount":2,"id":2,"item":"Mouse","user_id":2}]

C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost/orders/1
{"order":{"amount":1,"id":1,"item":"Laptop","user_id":1}, "user":{"email":"alice@example.com","id":1,"name":"Alice"}}
```